

PrimeSim™ HSPICE® User Guide: Demonstration Netlists

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About the PrimeSim HSPICE User Guide: Demonstration Netlists

The *PrimeSim HSPICE User Guide: Demonstration Netlists* provides detailed information on locating and running demonstration netlists, listing of demonstration input files and demonstration netlist examples.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeSim HSPICE Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeSim HSPICE tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Cadence® Virtuoso® Analog Design Environment

Synopsys Custom Compiler™

Synopsys WaveView™

Synopsys FineSim®

Synopsys CustomSim™

Synopsys SolvNetPlus support site

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

1

Locating and Running PrimeSim HSPICE Demonstration Netlists

Introduces the PrimeSim HSPICE demonstration netlists. Provides information on how to locate the demonstration files in your PrimeSim HSPICE installation and how to run the demonstrations.

The PrimeSim HSPICE tool is shipped with as many as 350 demonstration netlist files, under 25+ application categories. The demo files are designed as training examples to help you learn to construct and execute netlists to use PrimeSim HSPICE for various simulation applications.

This chapter discusses the following topics:

- [Locating the Demonstration Files](#)
- [Using the Demo Directory Tree](#)
- [Running the Demonstration Files](#)

Locating the Demonstration Files

Within your PrimeSim HSPICE installation, you can locate the demonstration files at:

- For PrimeSim HSPICE: `$installdir/demo/hspice/`
- For PrimeSim HSPICE advanced analyses: `$installdir/demo/hspice/rf_examples/`

Where `$installdir` is the directory in which, the PrimeSim HSPICE tool is installed on your network or machine.

[Chapter 2, Listing of Demonstration Input Files](#), of this guide provides a detailed listing of the PrimeSim HSPICE demonstration file categories, descriptions of the netlists, and the demonstration details.

Using the Demo Directory Tree

The tables in [Chapter 2, Listing of Demonstration Input Files](#), list the demonstration files. All PrimeSim HSPICE distributions include these examples in the demo directory tree under `$installdir/demo/hspice`.

Where `$installdir` is the directory in which, the PrimeSim HSPICE tool is installed on your network or machine.

[Table 1](#) lists the demo directory tree includes the following directories:

Table 1 *Demo Directories*

Directory	Description
/spiceADE_integ	PrimeSim HSPICE integration tutorial, Cadence Virtuoso Analog Design Environment
/apps	General applications
/back_annotation	Usage of BA_options
/behave	Analog behavioral components
/bench	Standard benchmarks
/bisection	Bisection optimization
/bjt	Bipolar components
/cchar	Characteristics of cell prototypes
/ciropt	Circuit level optimization
/devopt	Device level optimization
/encryption	Traditional, 8-bit, and 3DES encryption
/fft	Fourier analysis
/filters	Filters
/ibis	IBIS examples
/lstb	Loop stability analysis
/mag	Transformers, magnetic core components
/mos	MOS components

Table 1 Demo Directories (Continued)

Directory	Description
/mpo	Model Parameter Optimization
/rf_examples	RF examples
/si	Signal Integrity applications
/sources	Dependent and independent sources
/sparam	S-parameter applications
/tline	Filters and transmission lines
/twline	W-element transmission lines and field solvers
/variability	Variation Block, Monte Carlo, and AC/DC Mismatch examples
/veriloga	Verilog-A examples

Running the Demonstration Files

To run a demonstration file:

On Linux-based platforms:

1. Go to the directory where the demo file is located in the demo directory tree under `$installdir/demo/hspice`.

Where `$installdir` is the directory where the PrimeSim HSPICE tool is installed on your network or machine.

2. If you have set up the correct path to invoke and run PrimeSim HSPICE, execute the following command:

```
%> hspice -i your_selected_spice_demo_file -o output_file
```

On Windows platforms:

1. Open the PrimeSim HSPICE User Interface (HSPUI).

To open HSPUI, go to **Start > All Programs > Synopsys > hspice installed_version > hspui installed_version**.

2. In HSPUI, you can open a design file in one of the following ways:

- Select **File > Open** in the menu bar.
- Click the file folder icon in the icon bar.
- Click the **Browse** button next to the **Input netlist file** list.

For all the above cases, HSPUI opens a standard Windows file browser to help you select the design file for opening.

3. In the file browser, go to the directory where the demo file is located in the demo directory tree under `$installdir/demo/hspice`.

Where `$installdir` is the directory in, which the PrimeSim HSPICE tool is installed on your network or machine.

4. Click **File > Simulate** or click the Simulate icon in the icon bar.

2

Listing of Demonstration Input Files

Lists and describes the PrimeSim HSPICE demonstration input files; provides paths to these files in the general distribution.

The following groups of example demonstration files are available within your PrimeSim HSPICE installation:

- [PrimeSim HSPICE Integration to ADE Demonstration Examples](#)
- [Applications of General Interest Examples](#)
- [Back-Annotation Demo Cases](#)
- [Behavioral Application Examples](#)
- [Benchmark Examples](#)
- [Bisection-Timing Analysis Examples](#)
- [BJT and Diode Examples](#)
- [Cell Characterization Examples](#)
- [Circuit Optimization Examples](#)
- [Device Optimization Examples](#)
- [Encryption Examples](#)
- [Fourier Analysis Examples](#)
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- [IBIS Examples](#)
- [Loop Stability Analysis](#)
- [Magnetics Examples](#)
- [MOSFET Device Examples](#)
- [Model Parameter Optimization Examples](#)

- [RF Examples](#)
- [Signal Integrity Examples](#)
- [Sources Examples](#)
- [S-parameter Examples](#)
- [Transmission Lines Examples](#)
- [Transmission \(W-element\) Line Examples](#)
- [Variability Examples](#)
- [Exploration Block Examples](#)
- [Verilog-A Examples](#)

PrimeSim HSPICE Integration to ADE Demonstration Examples

Table 2 lists the PrimeSim HSPICE integration tutorial examples available in the `$installdir/demo/hspice/spiceADE_integ/` demo directory.

Table 2 PrimeSim HSPICE Integration Tutorial Examples

File Name	Description	Reference to Detailed Demo Description
Biaschk_Demo_61	.BIASCHK examples	See Dynamic Check Using the .BIASCHK Statement in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
Corner_Demo_51 Corner_Demo_61	Full example of corner analysis	See Running a Corner Analysis in PrimeSim HSPICE-ADE in the <i>HSPICE Integration to Cadence Virtuoso Analog Design Environment User Guide</i> .
Lstb_Demo_61	Loop stability analysis	See Using .LSTB for Loop Stability Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
Mixer_Demo_61	Netlisting and running of PrimeSim HSPICE advanced analog analyses (hb, hbac, hbnoise, sn, snac, and snnoise); PLL Demo_61 complements this demo	See Running an Advanced Analysis in PrimeSim HSPICE-ADE in the <i>HSPICE Integration to Cadence® Virtuoso® Analog Design Environment User Guide</i> .
Monte_Demo_51 Monte_Demo_61	Monte Carlo analysis	See Running a Monte Carlo Simulation in PrimeSim HSPICE-ADE in the <i>HSPICE Integration to Cadence Virtuoso Analog Design Environment User Guide</i> .
Mosra_Demo_51 Mosra_Demo_61	Full MOSFET reliability analysis	See MOSFET Reliability Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
Optimization_Demo_51 Optimization_Demo_61	Optimization analysis	See Running an Optimization Analysis in the <i>HSPICE Integration to Cadence Virtuoso Analog Design Environment User Guide</i> .
PLL_Demo_51 PLL_Demo_61	Suites of files for ADE versions 5.1xx and 6.1xx (required to run a guided tutorial in the first chapter of the <i>PrimeSim HSPICE Integration to Cadence® Virtuoso® Analog Design Environment User Guide</i> , including a Verilog-A example).	See Quick-Start Tutorial in the <i>HSPICE Integration to Cadence Virtuoso Analog Design Environment User Guide</i> .
RfMeas_Demo_61	RF examples	See Running PrimeSim HSPICE Advanced Analyses in the <i>HSPICE Integration to Cadence Virtuoso Analog Design Environment User Guide</i> .

Applications of General Interest Examples

Table 3 lists the applications of general interest examples available in the `$installdir/demo/hspice/apps/ demo` directory.

Table 3 *Applications of General Interest Examples*

File Name	Description	Reference to Detailed Demo Description
alm124.sp	AC, noise, and transient op-amp analysis	See Op-amp Characterization of LM124 in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
alter2.sp	.ALTER examples	See .DEL LIB with .ALTER in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
ampg.sp	Pole/zero analysis of a G source amplifier	See Pole/Zero Analysis Examples in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
asic1.sp	Ground bounce for I/O CMOS driver	See CMOS Output Driver Demo in this guide.
asic3.sp	Ten-stage lumped MOS model	See Modeling Wide-Channel MOS Transistors Demo in this guide.
biaschk.sp	Apply bias check analysis on D flip flop	See Dynamic Check Using the .BIASCHK Statement in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
bjtdiff.sp	BJT differential amp with every analysis type	
bjtschmt.sp	Bipolar Schmidt trigger	See .DC in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
bjtsense.sp	Bipolar sense amplifier	
cellchar.sp	Characteristics of ASIC inverter cell	See Performing Advanced Cell Characterization in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
four.sp	CMOS inverter applied with Fourier analysis	See Spectrum Analysis (Fourier Transform) in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
FFEdemo.sp	Feed-Forward Equalization with PrimeSim HSPICE	See Feed-Forward Equalization with the PrimeSim HSPICE Tool in the <i>PrimeSim HSPICE User Guide: Demonstration Netlists</i>
gaasamp.sp	Simple GaAsFET amplifier	
gen28.inc	Model library file	
grouptim.sp	Group time-delay example	

Table 3 Applications of General Interest Examples (Continued)

File Name	Description	Reference to Detailed Demo Description
inv.sp	Sweep MOSFET -3 sigma to +3 sigma use .MEASURE output	See Delay and Power Consumption Analysis of Inverters in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
mcdiff.sp	CMOS differential amplifier	See Example 4 – CMOS Differential Amplifier in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
• mondc_a.sp • mondc_b.sp	Monte Carlo of MOS diffusion and photolithographic effects Monte Carlo DC analysis	See Major and Minor Distribution in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
mont1.sp	Monte Carlo Gaussian, uniform, and limit function	See Gaussian, Uniform, and Limit Functions in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
mos2bit.sp	Two-bit MOS adder	See Two-Bit Adder Demo in this guide.
• noise_app.sp • noise_app_orig.sp	Uses the .LIN command to do a noise analysis Uses the .NOISE command on the same circuit as above	See Noise Simulation in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
quickAC.sp	AC analysis on a RC network	See AC Analysis of an RC Network in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
quickDC.sp	DC analysis on an inverter	See DC Analysis of an Inverter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
quickINV.sp	Transient analysis on an inverter	See Transient Analysis of an Inverter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
quickTRAN.sp	Transient analysis on a resistor divider	See Transient Analysis of an RC Network in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
rc_monte.sp	Monte Carlo analysis of RC Time Constant	See Monte Carlo Analysis of RC Time Constant in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
sclopass.sp	Switched-capacitor low-pass filter	See Modeling Switched Capacitor Filters in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
soa.sp	.biaschk example to enforce Safe Operating Area (SOA)	See Dynamic Check Using the .BIASCHK Statement in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
tf.sp	AC transfer function	

Table 3 Applications of General Interest Examples (Continued)

File Name	Description	Reference to Detailed Demo Description
tlib1	Model library file	
tlib2	Model library file	
tlib3	Model library file	
tlib4	Model library file	
trist_buf_opt.sp	Tri-State buffer optimization	See Optimizing CMOS Tristate Buffer in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
wildchar.sp	Wildcard print and probe example	See .PRINT and .PROBE in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
worst.sp	Worst-case skew models by using .ALTER	
xbjt2bit.sp	BJT NAND gate two-bit binary adder	See Two-Bit Adder Demo in this guide.
top.sp	Read-in a value change dump (VCD) file	See .VCD in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i>

Back-Annotation Demo Cases

Directories include: top-level netlists, extracted netlists, selected nets file, schematic level netlists, model files, and README files in the `$installdir/demo/hspice/back_annotation/` demo directory.

Table 4 Back-Annotation Cases

Case Name	Description	Reference to Detailed Description
option_ba_dpfpfx/	Demonstrates use of .OPTION BA_DPFPFX to prepend an extra prefix when searching the ideal netlist for instances referenced by the parasitic file.	See .OPTION BA_DPFPFX in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_file/dspf/multi ba/	Demonstrates use of the .OPTION BA_FILE command to launch multiple DSPF files for parasitic back-annotation.	See .OPTION BA_FILE in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_file/dspf/	Demonstrates use of the .OPTION BA_FILE command for single DSPF file.	See .OPTION BA_FILE in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .

Table 4 Back-Annotation Cases (Continued)

Case Name	Description	Reference to Detailed Description
option_ba_file/spef/	Demonstrates use of the .OPTION BA_FILE command for single SPEF file.	See .OPTION BA_FILE in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_fingerdelim/	Demonstrates use of .OPTION BA_FINGERDELIM to explicitly specify the delimiter character used for finger devices.	See .OPTION BA_FINGERDELIM in the <i>PrimeSim™ Continuum Reference Manual: Commands and Control Options</i> .
option_ba_hierdelim/	Demonstrates use of .OPTION BA_HIERDELIM to specify the hierarchical separator in the DSPF file.	See .OPTION BA_HIERDELIM in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_idealpfx/	Demonstrates use of .OPTION BA_IDEALPFX to have the simulator add a second prepended prefix when doing a search of the ideal netlist.	See .OPTION BA_IDEALPFX in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_mergeport/	Demonstrates use of .OPTION BA_MERGEPORT to control whether to merge net ports into one node.	See .OPTION BA_MERGEPORT in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_print/	Demonstrates use of .OPTION BA_PRINT to control whether to output nodes and resistors/capacitors introduced by back-annotation.	See .OPTION BA_PRINT in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
option_ba_terminal/	Demonstrates use of .OPTION BA_TERMINAL to specify the terminal name mapping between the parasitic netlist and the terminal names recognized by the simulator.	See .OPTION BA_TERMINAL in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
selective_ba/ba_acheck/	Demonstrates the use of the .BA_ACHECK command to specify the rule for detecting node activity in back-annotation.	See .BA_ACHECK in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
selective_ba/option_ba_active/	Demonstrates use of .OPTION BA_ACTIVE to create an Active Net file for selective back-annotation.	See .OPTION BA_ACTIVE in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
selective_ba/option_ba_activehier/	Demonstrates use of .OPTION BA_ACTIVEHIER to annotate full hierarchical net names that are specified for BA_ACTIVE files.	See .OPTION BA_ACTIVEHIER in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .
selective_ba/option_ba_coupling/	Demonstrates use of .OPTION BA_COUPLING to control how to treat cutoff coupling capacitors when invoking selective net back-annotation.	See .OPTION BA_COUPLING in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .

Table 4 Back-Annotation Cases (Continued)

Case Name	Description	Reference to Detailed Description
selective_ba/option_ba_netfmt/	Demonstrates use of .OPTION BA_NETFMT to specify the format of the Active Net file (*.rcxt or *.hsimba).	See .OPTION BA_NETFMT in the <i>PrimeSim Continuum Reference Manual: Commands and Control Options</i> .

Behavioral Application Examples

Table 5 lists the applications of behavioral application examples available in the \$installdir/demo/hspice/behave/ demo directory.

Table 5 Behavioral Application Examples

File Name	Description	Reference to Detailed Description
acl.sp	Acl gate	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
amp_mod.sp	Amplitude modulator with pulse waveform carrier	See Behavioral Amplitude Modulator in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
behave.sp	AND/NAND gates by using G-, E-elements	See Behavioral AND and NAND Gates in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> . See also G-element Examples and E-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
calg2.sp	Voltage variable capacitance	See LC Oscillator in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
compar.sp	Behavioral comparator with hysteresis	See Inverting Comparator with Hysteresis in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
det_dff.sp	Double edge-triggered flip-flop	See Behavioral Double-Edge Triggered Flip-Flop in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> . See also G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
diff.sp	Differentiator amplifier and opamp signals	See Behavioral Differentiator in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
diode.sp	Behavioral diode by using a Piecewise Linear (PWL) VCCS	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .

Table 5 Behavioral Application Examples (Continued)

File Name	Description	Reference to Detailed Description
dlatch.sp	CMOS D-latch by using behavioral elements	See Behavioral D-Latch in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
galg1.sp	Sampling a sine wave	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
idealop.sp	Ninth-order low-pass filter	See E-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
integ.sp	Integrator circuit	See Behavioral Integrator in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
inv_vin_vout.sp	DC sweep of an INV	See Creating a Behavioral Inverter Lookup Table in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
invb_op.sp	Optimizes the CMOS macromodel inverter	See Optimizing Behavioral CMOS Inverters in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
ivx.sp	Characteristics of the PMOS and NMOS as a switch	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
op_amp.sp	Op-amp from Chua and Lin	See 741 Op-Amp from Controlled Sources in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
pdb.sp	Phase detector by using behavioral NAND gates	See Phase Detector with Multi-Input NAND Gates in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
- pll.sp - pll_bvp.sp	- PLL build with BJT - PLL build with behavioral source	See Phase Locked Loop Modeling in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
- pwl2.sp - pwl4.sp	PPW-VCCS with a gain of 1 amp/volt Eight-input NAND gate	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
pwl7.sp	Modeling inverter by using a PWL VCVS	See E-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
pwl8.sp	Smoothing the triangle waveform by using the PWL Current-controlled Current Source (CCCS)	See F-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
pwl10.sp	Operational amplifier used as a voltage follower	See H-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
ring5bm.sp	Five-stage ring oscillator – macromodel CMOS inverter	See Optimizing Behavioral Ring Oscillators in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .

Table 5 Behavioral Application Examples (Continued)

File Name	Description	Reference to Detailed Description
ringb.sp	Ring oscillator by using behavioral model	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
rtest.sp	Voltage-controlled resistor, inverter chain	
sampling.sp	Sampling a sine wave	See Behavioral Data Sampler in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
swcap5.sp	Fifth-order elliptic switched capacitor filter	See Switched Capacitor Filter Example in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
switch.sp	Test for PWL switch element	See G-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
swrc.sp	Switched capacitor RC circuit	
vcob.sp	Voltage-controlled oscillator by using PWL functions	See Voltage-Controlled Oscillator in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .

Benchmark Examples

Table 6 lists the applications of benchmark examples available in the `$installdir/demo/hspice/bench/` demo directory.

Table 6 Benchmark Examples

File Name	Description	Reference to Detailed Demo Description
bigmos1.sp	Large MOS simulation	
demo.sp	Quick demo file to test installation	See Invoking PrimeSim HSPICE Simulations in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
digstim.vec	Vector stimulus file for m2bit_v.sp	See Specifying a Digital Vector File and Mixed Mode Stimuli in the <i>PrimeSim Continuum User Guide: Elements</i> .
example.sp	CMOS amplifier	
m2bit.sp	72-transistor two-bit adder – typical cell simulation	
m2bit_v.sp	Same as m2bit.sp except using vector stimulus file	
senseamp.sp	Bipolar analog test case	

Bisection-Timing Analysis Examples

Table 7 lists the applications of bisection-timing examples available in the `$installdir/demo/hspice/bisect/` demo directory.

Table 7 Bisection-Timing Examples

File Name	Description	Reference to Detailed Demo Description
dff_push.sp	DFF pushout bisection search for setup time	See Pushout Bisection Methodology in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
dff_top.sp	DFF bisection search for setup time	See Setup Time Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
fig26_4.sp	Early, optimal and late setup times of DFF	See Overview of Bisection in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
inv_a.sp	Inverter bisection (pass-fail)	See Minimum Pulse Width Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
tsmc018.m	TSMC model file used by dff_push.sp	

BJT and Diode Examples

Table 8 lists the applications of BJT and Diode examples available in the `$installdir/demo/hspice/bjt/` demo directory.

Table 8 BJT and Diode Examples

File Name	Description	Reference to Detailed Demo Description
bjtbeta.sp	plot BJT beta	See BJT Models in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
bjtgms.sp	plot BJT Gm, Gpi	
dpntun.sp	junction tunnel diode	
hicum.sp	HICUM BJT MOS terminal characterization	See LEVEL=8 HICUM Model in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
mextram.sp	I-V characteristics of a MEXTRAM BJT	See LEVEL=6 Philips Bipolar Model (MEXTRAM) in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
mextram_ac.sp	AC analysis of a MEXTRAM BJT	

Table 8 BJT and Diode Examples (Continued)

File Name	Description	Reference to Detailed Demo Description
mextram_dc.sp	DC analysis of a MEXTRAM BJT	
mextram_tran.sp	Tran analysis of a MEXTRAM BJT	
quasisat.sp	Comparison of BJT level1 and level2	See BJT Quasi-Saturation Model in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
self-heat.sp	VBIC BJT with self heating feature	See VBIC Bipolar Transistor Model in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
vbic.sp	DC of a VBIC BJT	
vbic99_ac.sp	NET analysis of a VBIC99 BJT	See LEVEL=9 VBIC99 Model in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
vbic99_dc.sp	DC analysis of a VBIC99 BJT	
vbic99_tran.sp	Transient analysis of a VBIC99 BJT	

Cell Characterization Examples

Table 9 lists the applications of cell characterization examples available in the `$installdir/demo/hspice/cchar/` demo directory.

Table 9 Cell Characterization Examples

File Name	Description	Reference to Detailed Demo Description
diff.sp	.Model opt, hemocytoblastic	
gen28.inc	Level 28 model library used by netlists	
inv3.sp	The inv3.sp file is the characteristics of an inverter, .model opt, method=passfail	
inva.sp	Characteristics of an inverter, .model opt, method=passfail	
invb.sp	Characteristics of an inverter, .model opt, method=bisection	
load1.sp	Inverter sweep, delay versus fanout	See Delay versus Fanout in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
setupbsc.sp	Setup characteristics	
setupold.sp	Setup characteristics, .model opt	

Table 9 Cell Characterization Examples (Continued)

File Name	Description	Reference to Detailed Demo Description
setuppas.sp	Setup characteristics, .model opt, method=passfail	
sigma.sp	The sigma.sp file sweep MOSFET -3 sigma to +3 sigma by using measure output	See Rise, Fall, and Delay Calculations in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Circuit Optimization Examples

Table 10 lists the applications of circuit optimization examples available in the `$installdir/demo/hspice/ciropt/` demo directory.

Table 10 Circuit Optimization Examples

File Name	Description	Reference to Detailed Demo Description
ampgain.sp	Set unity gain frequency of a BJT differential pair	See FIND and WHEN Functions in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
ampopt.sp	Optimize area, power, and speed of a MOS amp	See Optimizing MOS Op-amp in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
asic2.sp	Optimize speed or power of a CMOS output buffer	
asic6.sp	Find best width of a CMOS input buffer	
delayopt.sp	Optimize group delay of an LCR circuit	
lpopt.sp	Match lossy filter to ideal filter	
opttemp.sp	Find first and second temperature coefficients of a resistor	See Temperature Coefficients Demo in this guide.
rcopt.sp	Optimize speed or power for an RC circuit	See RC Network Optimization in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Device Optimization Examples

Table 11 lists the applications of device optimization examples available in the `$installdir/demo/hspice/devopt/` demo directory.

Table 11 *Device Optimization*

File name	Description	Reference to Detailed Demo Description
beta.sp	LEVEL=2 beta optimization	
bjtopt.sp	S-parameter optimization of a 2n6604 BJT	See BJT S-parameters Optimization in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
bjtopt1.sp	2n2222 DC optimization	
bjtopt2.sp	2n2222 Hfe optimization	
d.sp	Diode, multiple temperatures	
dcopt1.sp	1n3019 diode, I-V and C-V optimization	
jopt.sp	300u/1u GaAs FET, DC optimization	See Optimizing GaAsFET Model DC in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
ml13opt.sp	MOS LEVEL=13 DC optimization	See MOS Level 13 Model DC Optimization in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
ml2opt.sp	MOS LEVEL=3 I-V optimization	
opt_bjt.sp	T2N9547 BJT Optimization	See BJT Model DC Optimization in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Encryption Examples

[Table 12](#) lists the applications of device encryption examples available in the `$installdir/demo/hspice/encryption/` demo directory.

Table 12 *Encryption Examples*

File name	Description	Reference to Detailed Demo Description
8-byte_key.tar	Suite of files demonstrating how to set up an 8-byte_key encryption file.	See 8-Byte Key Encryption in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
traditional.tar	Suite of files demonstrating how to set up a traditional (free_lib) encryption file.	See Traditional Library Encryption in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Table 12 *Encryption Examples (Continued)*

File name	Description	Reference to Detailed Demo Description
triple_DES.tar	Suite of files plus a directory of 2 lib_DES libraries demonstrating how to set up a 3DES encryption file.	See Triple DES (3DES) Public and Random Keys in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
aes256_encrypt/	Suites of files and sub-directories demonstrating how to set up an AES256 encryption file.	See AES256 Encryption in the <i>PrimeSim Continuum Overview Guide</i> , Library and Data Encryption.
enc2023_encrypt/	Suites of files and sub-directories demonstrating how to set up an ENC2023 encryption file.	See ENC2023 Encryption in the <i>PrimeSim Continuum Overview Guide</i> , Library and Data Encryption.
expire_date	Suites of files and sub-directories demonstrating how to set up an encryption file with expired date.	See Expire_date encryption in the <i>PrimeSim Continuum Overview Guide</i> , Library and Data Encryption.
demo.lib		
design.sp		
dff.sp		
README and auxiliary files		

Fourier Analysis Examples

[Table 13](#) lists the applications of fast fourier transform examples available in the `$installdir/demo/hspice/fft/ demo` directory.

Table 13 *Fast Fourier Transform Examples*

File Name	Description	Reference to Detailed Demo Description
am.sp	Amplitude modulation	
balance.sp	Balanced modulator/demodulator	See Balanced Modulator and Demodulator in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
bart.sp	Bartlet window	See Using Windows in FFT Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
black.sp	Blackman window	
dist.sp	Second harmonic distortion	See Spectrum Analysis (Fourier Transform) in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Table 13 *Fast Fourier Transform Examples (Continued)*

File Name	Description	Reference to Detailed Demo Description
exam1.sp	AM modulation	See AM Modulation in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
exam3.sp	Mixer signal detection	See Signal Detection Test Circuit in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
exam4.sp	Harmonic distortion	See Spectrum Analysis (Fourier Transform) in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
exp.sp	Exponential source	See Exponential Source Function in the <i>PrimeSim™ Continuum User Guide: Elements</i> .
fft.sp	Sine FFT test	See Spectrum Analysis (Fourier Transform) in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
fft1.sp	FFT test 1	
fft2.sp	FFT test 2	
fft3.sp	FFT test 3	
fft4.sp	FFT test 4	See Using Windows in FFT Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
fft5.sp	FFT analysis, data-driven transient analysis	
fft6.sp	FFT analysis, sinusoidal source	
gauss.sp	FFT analysis, Gaussian window	
hamm.sp	FFT analysis, Hamming window	See Trapezoidal Pulse Source in the <i>PrimeSim HSPICE User Guide: Elements</i> .
hann.sp	FFT analysis, Hanning window	
harris.sp	FFT analysis, Blackman-Harris window	
intermod.sp	FFT analysis, intermodulation distortion	
kaiser.sp	FFT analysis, Kaiser window	See PWL Source in the <i>PrimeSim HSPICE User Guide: Elements</i> .
mod.sp	FFT analysis, modulated pulse	
pulse.sp	FFT analysis, pulse source	
pwl.sp	FFT analysis, piecewise linear source	
rect.sp	FFT analysis, rectangular window	See Using Windows in FFT Analysis in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
rectan.sp	FFT analysis, rectangular window	

Table 13 Fast Fourier Transform Examples (Continued)

File Name	Description	Reference to Detailed Demo Description
sffm.sp	FFT analysis, single-frequency FM source	See Single-frequency FM Source in the <i>PrimeSim HSPICE User Guide: Elements</i> .
sine.sp	FFT analysis, sinusoidal source	See Spectrum Analysis (Fourier Transform) in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
swcap5.sp	FFT analysis, fifth-order elliptic, switched-capacitor filter	
tri.sp	FFT analysis, rectangular window	
win.sp	FFT analysis, window test	
window.sp	FFT analysis, window test	
winreal.sp	FFT analysis, window test	

Filters Examples

Table 14 lists the applications of filters examples available in the `$installdir/demo/hspice/filters/` demo directory.

Table 14 Filters

File Name	Description	Reference to Detailed Demo Description
bandstopl.sp	Band reject filter, AC and transient analysis	See Laplace Band-Reject Filter in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
fbp_1.sp	Bandpass LCR filter, measurement	
fbp_2.sp	Bandpass LCR filter, pole/zero	
fbprlc.sp	LCR AC analysis for resonance	
fhp4th.sp	High-pass LCR, fourth-order Butterworth filter, pole-zero analysis	See Example 3 – High-Pass Butterworth Filter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
fkerwin.sp	Pole/zero analysis of Kerwin's circuit	See Example 2 – Kerwin's Circuit in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
flp5th.sp	Low-pass, fifth-order filter, pole-zero analysis	See Example 1 – Low-Pass Filter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Table 14 *Filters (Continued)*

File Name	Description	Reference to Detailed Demo Description
flp9th.sp	Low-pass, ninth-order FNDNR, with ideal op-amps, pole-zero analysis	See Example 6 – Active Low-Pass Filter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
lcline.sp	LC line model using Laplace behavioral elements	See Y-Parameter Line Modeling in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
low_pass.sp	Behavioral model using E and G elements	See Laplace Low-Pass Filter in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
low_pass9a.sp	Active low-pass filter using behavioral opamp models	
lowloss.sp	RL line model using Laplace behavioral elements	See AWE Transfer Function Modeling in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
ninth.sp	Active low-pass filter using Laplace elements	
phaseshift.sp	Behavioral model using G table element	See Circular Convolution Example in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
rcline.sp	RC line model using Laplace elements	See RC Line Modeling in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .

Note:

See also `fbplin.sp` in `$installdir/demo/hspice/sparam` for an LC band pass filter example with `.LIN` analysis.

IBIS Examples

[Table 15](#) lists the applications of IBIS modeling files examples available in the `$installdir/demo/hspice/ibis/` demo directory.

Table 15 *IBIS Modeling Files*

File name	Description	Reference to Detailed Demo Description
at16245.ibs	IBIS model file, used in <code>iob_ex1.sp</code> example file	See Modeling Input/Output Buffers Using IBIS Files in the <i>PrimeSim Continuum User Guide: Elements</i> .

Table 15 *IBIS Modeling Files (Continued)*

File name	Description	Reference to Detailed Demo Description
iob_ex1.sp	Using IBIS buffer example	
ebd	EBD directory	
ebd/cmpt1.ibs	IBIS model file	
ebd/ebd.ebd	IBIS EBD file example	
ebd/ebd.sp	Using EBD files example	
ebd/pinmap.ebd	IBIS EBD file example, uses pin mapping	
ebd/pinmap.ibs	IBIS model file	
ebd/pinmap.sp	Using EBD files example	
icm	ICM directory	
icm/readme	Readme file for ICM examples	
icm/nodepath_rl gc/bga_1.sp	Using ICM with nodepath description example	
icm/nodepath_rl gc/bga_example .icm	ICM example file	
icm/nodepath_s ele/s_w_test_G Hz_db.s4p	TouchStone file for ICM example, called by test1.icm	
icm/nodepath_s ele/sect2_s_2. inc	S-parameter model call, used by test1.sp	
icm/nodepath_s ele/test1.icm	ICM example file	
icm/nodepath_s ele/test1.sp	Using ICM with nodepath description and S-element	
icm/treepath_rl gc/sect3_rlgc_4. inc	RLGC file used by test1.sp	
icm/treepath_rl gc/sect_w_4.inc	RLGC file used by test1.sp	
icm/treepath_rl gc/test1.icm	ICM example file	

Table 15 *IBIS Modeling Files (Continued)*

File name	Description	Reference to Detailed Demo Description
icm/treepath_rl gc/test1.sp	Using ICM with treepath and rlgc data example	
icm/treepath_sw ath/complex.icm	ICM example file	
icm/treepath_sw ath/complex.sp	Using ICM with swath matrix expansion	
schedule	Driver schedule directory	
schedule.ibs		
schedule.sp		
ibis5.0	IBIS5.0 directory	
ibis5.0/README		
ibis5.0/test4p2 .sp		
ibis5.0/test4p2_s nps.ibs		
ibis5.0/test5p0 .sp		
ibis5.0/test5p0_s nps.ibs		
external_circuit	Directory with example on IBIS external models	
external_circuit/t est_new.sp		

Loop Stability Analysis

Table 16 lists the applications of loop stability examples available in the `$installdir/demo/hspice/lstb/` demo directory.

Table 16 *Loop Stability*

File Name	Description	Reference to Detailed Demo Description
single.sp	.LSTB single-ended mode example for ideal inverting amplifier with single pole	See Single-Ended Mode Example: Ideal Inverter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Magnetics Examples

Table 17 lists the applications of magnetic examples available in the `$installdir/demo/hspice/mag/` demo directory.

Table 17 *Magnetics*

File Name	Description	Reference to Detailed Demo Description
aircore.sp	Air-core transformer circuit	See Inductor Device Model and Equations in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
bhloop.sp	Magnetic core model, plot B-H loop characteristics	See Inductor Core Models in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
jiles.sp	Effects of core model parameters on B-H loop characteristics	See Jiles-Atherton Ferromagnetic Core Model in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
magcore.sp	Magnetic-core transformer circuit	
tj_opt.sp	Optimizing magnetic core parameters	See Optimizing the Extraction of Parameters in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .
tj2b.sp	Hysteresis effects in magnetic cores	See Discontinuities in Inductance Due to Hysteresis in the <i>PrimeSim Continuum Reference Manual: Device Models</i> .

MOSFET Device Examples

Table 18 lists the applications of Mosfet device examples available in the `$installdir/demo/hspice/mos/` demo directory.

Table 18 *MOSFET Devices*

File Name	Description	Reference to Detailed Demo Description
calcap.sp	Calculate AC gate capacitance	See Calculating Gate Capacitance in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
calcap.results	Results file from calcap.sp	
capop0.sp	Plot MOS capacitances, LEVEL=2	See CAPOP=0 – SPICE Meyer Gate Capacitances in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
capop1.sp	Plot MOS capacitances, LEVEL=2	See CAPOP=1 – Modified Meyer Gate Capacitances in the <i>PrimeSim HSPICE Reference Manual: MOSFET Models</i> .

Table 18 *MOSFET Devices (Continued)*

File Name	Description	Reference to Detailed Demo Description
capop2.sp	Plot MOS capacitances, LEVEL=2	See CAPOP=2 – Parameterized Modified Meyer Capacitance in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
cascode.sp	MOS Cascode amplifier example. Shows effect of level=3 impact ionization parameter.	See CAPOP=3 – Gate Capacitances (Simpson Integration) in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
chrgpump.sp	Charge-conservation test. Charges pump using LEVEL=3 MOS.	See CAPOP=4 – Charge Conservation Capacitance Model in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
gatecap.sp	DC gate capacitance calculation	See Gate Capacitance – Plotting Gate Capacitances in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
mcap2_a.sp	MOS charge conservation capacitances	See CAPOP=4 – Charge Conservation Capacitance Model in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
mcap3.sp	MOS charge conservation capacitances	See CAPOP=4 – Charge Conservation Capacitance Model in the <i>PrimeSim HSPICE Reference Manual: MOSFET Models</i> .
ml5iv.sp	MOS LEVEL=5 example	See LEVEL 5: AMI-ASPEC Depletion and Enhancement (Taylor-Huang) Model in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
ml13iv.sp	Plot I-V for LEVEL=13	See BSIM Model – Compatibility Notes: IDS and VGS Curves for PMOS and NMOS in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
ml13opt.sp	Optimizing MOS LEVEL=13 model parameter	See MOS Level 13 Model DC Optimization in the <i>PrimeSim Continuum User Guide: Basic Simulation and Analysis</i> .
ml27iv.sp	Plot I-V for LEVEL=27 SOSFET	See LEVEL 27: SOSFET Model in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
mosiv.sp	Plot I-V for files that you include	
mosivcv.sp	Example of plotting I-V and C-V curves, uses LEVEL=3 model	See MOS I-V and C-V Plotting Demo in this guide.
nch0.inc	MOS model for mosiv.sp and cap_m.sp	
selector.sp	Automatic model selector for width and length	See Searching Models as Function of W, L in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .

Table 18 MOSFET Devices (Continued)

File Name	Description	Reference to Detailed Demo Description
ssoi.sp	Floating bulk model	See SOSFET Model — Non-Fully Depleted SOI Model in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
t1.sp	MOS LEVEL=13 TOX calculation test	See BSIM Model Calculations in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
tempdep.sp	MOS LEVEL=3 temperature dependence	See MOS3 Empirical Model – Temperature Compensation Example in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .
tgam2.sp	LEVEL=6, gamma model	See ASPEC Model, (UPDATE=0 or 2) – Improved Multi-Level Gamma, UPDATE=1 in the <i>PrimeSim Continuum Reference Manual: MOSFET Models</i> .

Model Parameter Optimization Examples

Table 19 lists the applications of Model Parameter Optimization (MPO) examples available in the `$installdir/demo/hspice/mpo/` demo directory.

Table 19 MPO Examples

File Name	Description	Reference to Detailed Demo Description
test.sp	SPICE netlist	Example 1: IV Data Fitting
target.dat	Target data file	
target.txt	Target data integration file	
param.txt	Parameter file	
config.json	Configuration file	
test_mpo.py	mpo usage example	
readme	Demo case usage guide	

RF Examples

Table 20 lists the RF examples available in the `$installdir/demo/hspice/rf_examples/` demo directory.

Table 20 RF Examples

File Name	Description	Reference to Detailed Demo Description
acpr.sp	Envelope simulation example	See Envelope Simulation in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
bjt.inc	Transistor model library used by osc.sp	
clockbuf4.sp	Transient noise analysis example	See Example: Measure TIE Jitter and Jitter Spectrum in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
clockbuf4sn.sp	Shooting Newton-based analysis of clock buffer circuit example	See Shooting Newton Analysis – Tutorial Example in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
cmos49_model.inc	Transistor model library used by example circuits	
cmos90nmWflicker.lib	Transistor model library used by phasefreqdet.sp	
gpsvco.sp	Oscillator and Phase Noise analysis example	See Using HBOSC for a CMOS GPS VCO in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
gsmlna.sp	LNA Linear analysis example	See .LIN Analysis for a NMOS Low Noise Amplifier in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
gsmlnaIP3_A.sp	3rd order intercept point example	See Using HB Analysis for a Low Noise Amplifier in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
mix_hb.sp	Mixer HB analysis example	See Two-Tone HB Analysis in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
mix_hbac.sp	Mixer HBAC analysis example	See HBAC Analysis of a Simple Mixer in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
mix_snac.sp	Mixer Shooting Newton AC example	See SNAC Analysis Example in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
mix_tran.sp	Mixer transient analysis example	See Mixer Example: Transient Analysis in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .

Table 20 RF Examples (Continued)

File Name	Description	Reference to Detailed Demo Description
osc.sp	Oscillator tuning curve and phase noise analysis example	See HBOSC Analysis: Colpitts Oscillator in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
pa.sp	Power amplifier HB analysis example	See Using HB Analysis for a Power Amplifier in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
pfdcGain.sp	Shooting Newton analysis example	See Shooting Newton Analysis in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
phasefreqdet.sp	Shooting Newton and noise analysis example	See Shooting Newton Noise Analysis in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
rcInvPTDnoise.sp	Periodic Time-Dependent Noise analysis (.PTDNOISE) example	See Periodic Time-Dependent Noise Analysis in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
ringoscSN.sp	Shooting Newton and Phase Noise analysis example	See Ring Oscillator Example in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .
tsmc018.m	Transistor model library used by ringoscSN.sp	

Signal Integrity Examples

Table 21 lists the applications of signal integrity available in the `$installdir/demo/hspice/si/` demo directory.

See also [IBIS Examples](#), [S-parameter Examples](#), [Transmission Lines Examples](#), and [Transmission \(W-element\) Line Examples](#).

Table 21 Signal Integrity

File Name	Description	Reference to Detailed Demo Description
iotran.sp	Signetics I/O buffer with transmission lines example	See Simulating Circuits with Signetics Drivers in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
ipopt.sp	Time domain reflectometry (TDR) Optimization Example	
qa8.sp	Xilinx I/O buffer with transmission lines example	See Coupled Noise Simulation in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .

Table 21 Signal Integrity (Continued)

File Name	Description	Reference to Detailed Demo Description
qabounce.sp	Ground bounce example	See Ground-Bounce Simulation in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
stateye_ex1.sp	StatEye example	See Statistical Eye Analysis Setup in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
stateye_unfold.sp	StatEye unfolding eye diagram example	See Unfolding Statistical Eye Diagrams to Waveforms in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
AMI_Demo	IBIS-AMI demo example	See StatEye Analysis with AMI in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
PAM4	Directory containing the following examples for PAM4 source usage: - steTran.sp: PAM4 source with StatEye analysis - tran.sp: PAM4 source with normal transient analysis	See Pattern Source in the <i>PrimeSim HSPICE User Guide: Elements</i> .
SNPS_AMI	SNPS_AMI directory and files	
sputil	SPUTIL examples	See S-Parameter Standalone Manipulation Utility in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
stateyeAMI	StatEye example with AMI	See StatEye Analysis with AMI in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
nUI_sj.sp /hspice/StatEye_clk_skenw/nUI_sj.sp	N-Cycle Jitter Modeling for Data and Clock Signals	See N-Cycle Jitter Modeling for Data and Clock Signals in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
make_ssn.sp	Create SSN eye diagram example	See Capturing Steady State Noise for Power Integrity in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
add_ssn.sp	Add SSN to the eye diagram example	See Capturing Steady State Noise for Power Integrity in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
ste_wdf.sp	StatEye w/ pre-computed transient waveform example	See Statistical Eye Full Transient Mode in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .

Table 21 Signal Integrity (Continued)

File Name	Description	Reference to Detailed Demo Description
Xt_meas.sp	StatEye crosstalk using W element example	See Statistical Eye Analysis Setup in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .

Sources Examples

Table 22 lists the applications of source examples available in the `$installdir/demo/hspice/sources/` demo directory.

Table 22 Sources

File Name	Description	Reference to Detailed Demo Description
amsrc.sp	Amplitude modulation source example	See Single-frequency AM Source in the <i>PrimeSim Continuum User Guide: Elements</i> .
datamanual_pwl.sp	Simple data pwl example that does not use any sweep or tran tstep/stop syntax. The results are individual voltage waveforms.	See PWL Source in the <i>PrimeSim HSPICE User Guide: Elements</i> .
eelm.sp	E-element AC source example	See Using the E-element for AC Analysis in the <i>PrimeSim™ Continuum User Guide: Elements</i> .
exp.sp	Exponential independent source example	See Exponential Source Function in the <i>PrimeSim Continuum User Guide: Elements</i> .
prbs.sp	PRBS source example	See Pseudo Random Bit Generator Source in the <i>PrimeSim Continuum User Guide: Elements</i> .
pulse.sp	Pulse source example	See Trapezoidal Pulse Source in the <i>PrimeSim Continuum User Guide: Elements</i> .
pwl.sp	Repeated PWL source example	See PWL Source in the <i>PrimeSim Continuum User Guide: Elements</i> .
sffm.sp	Single-frequency, FM modulation source example	See Single-frequency FM Source in the <i>PrimeSim Continuum User Guide: Elements</i> .
sin.sp	Sinusoidal source, waveform example	See Sinusoidal Source Function in the <i>PrimeSim Continuum User Guide: Elements</i> .
supplynoise.sp	Example of a noise generator used for a pulse or DC level	See Example: Noise Generator Used for a Pulse or DC Level in the <i>PrimeSim Continuum User Guide: Elements</i> .
uelm.d2a	Part of uelem.sp example	

Table 22 Sources (Continued)

File Name	Description	Reference to Detailed Demo Description
uelm.sp	Digital U-element source example	See U-element Examples in the <i>PrimeSim Continuum User Guide: Elements</i> .
vcr1.sp	Switched-capacitor network by using G-switch	See Switched Capacitor Filter Example in the <i>PrimeSim HSPICE User Guide: Advanced Analog Simulation and Analysis</i> .

S-parameter Examples

Table 23 lists the applications of S-Parameter examples available in the `$installdir/demo/hspice/sparam/` demo directory.

Table 23 S-Parameter Examples

File Name	Description	Reference to Detailed Demo Description
diffamp_s.sp	Mixed-mode S-parameter example, differential amplifier; Port element declaration. S-element with mixed mode. The format is FQMODEL.	See Using the Mixed-Mode S-Parameters in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
fbplin.sp	An LC bandpass filter example with .LIN analysis.	See Extracting S-Parameters for a Bandpass Filter in the <i>PrimeSim HSPICE User Guide: Signal Integrity Modeling and Analysis</i> .
mixed2p.s4p	Port element declaration. S-element with mixed mode. The parameter format is Touchstone.	See Port Element in the <i>PrimeSim Continuum User Guide: Elements</i> .
mixedmode_s.sp	Mixed mode S-parameter example, transmission line	See Using the Mixed-Mode S-Parameters in the <i>PrimeSim Continuum User Guide: Signal Integrity Modeling and Analysis</i> .
sparam.sp	Using S-parameter model in SP model format	See S-element in the <i>PrimeSim Continuum User Guide: Elements</i> .
spciti.sp	S-element example, calling CITI format S-parameter file	
spmod.sp	S-element example, calling Touchstone format S-parameter file	
ss_citi.citi	CITI format S-parameter file example	
ss_ts.s2p	Touchstone format S-parameter file example	

Transmission Lines Examples

[Table 24](#) lists the applications of transmission lines (tline) examples available in the `$installdir/demo/hspice/tline/` demo directory.

Table 24 *Transmission Lines (tline) Example Files*

File name	Description	Reference to Detailed Demo Description
rcfilt.inc	RC filter macro model	
strip1.sp	U-element, two microstrips, in series (8 mil and 16 mil wide)	
strip2.sp	U-element, two microstrips, coupled together	
stripline.sp	U-element strip line example	See Three Coupled Lines, Stripline Configuration in the <i>PrimeSim Continuum User Guide: Elements</i> .
uele.sp	U-element, three coupled lines	See Three Coupled Lines, Sea of Dielectric Configuration in the <i>PrimeSim Continuum User Guide: Elements</i> .

Transmission (W-element) Line Examples

[Table 25](#) lists the applications of Twline demo files available in the `$installdir/demo/hspice/twline/` demo directory.

Table 25 *Twline Demo Files*

File Name	Description	Reference to Detailed Description
ex1.sp	4-conductor RLGC model W-element example	See W-element (Distributed Transmission Lines) in the <i>PrimeSim HSPICE User Guide: Elements</i> .
ex2.sp	4-conductor RLGC file W-element example	
ex3.sp	4-conductor W-element using U-element parameters	
example.rlc	RLGC file used by ex2.sp	
fs_ex1.sp	Field solver, conductor above ground plane	
fs_ex2.sp	Field solver, three-trace example	

Table 25 Twline Demo Files (Continued)

File Name	Description	Reference to Detailed Description
fs_ex3.sp	Field solver, coupled line example	
fs_ex4.sp	Field solver, Monte Carlo example	
petl_ex1.sp	Field solver, 1-conductor coax example	
petl_ex2.sp	Field solver, 2-conductor coax example	
rlgc.rlc	RLGC file used by rlgc.sp	
rlgc.sp	W-element using RLGC file	
umodel.sp	4-conductor W-element using U-element parameters	

Variability Examples

Table 26 lists the applications of variation block, Monte Carlo, and mismatching demo files available in the `$installdir/demo/hspice/variability/` demo directory.

Table 26 Variation Block, Monte Carlo, and Mismatching Demo Files

File name	Description	Reference to Detailed Description
EB_definition.txt	Exploration block definition text file	See Variation Block Examples in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
EB_explore.sp	Exploration block example 1	
EB_export.sp	Exploration block example 2	
matrix.sp	Matrix of 9 resistors for testing spatial variation with Monte Carlo	
monte_test.tar	Suite of DC test files named <code>test1.sp</code> through <code>test20.sp</code> to test combinations of resistors, subckts, model/instance parameters, and so on.	See Demos for Monte Carlo in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
opampacm.sp	Operational amplifier for ACMatch testing with Variation Block	See ACMatch Analysis Examples and Variation Block Examples in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
opampdcm.sp	Operational amplifier for DCMatch testing	See DCMatch Example in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .

Table 26 Variation Block, Monte Carlo, and Mismatching Demo Files (Continued)

File name	Description	Reference to Detailed Description
opampmc.sp	Operational amplifier for Monte Carlo testing with Variation Block	See Variation Block Examples in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
README and library files		

Exploration Block Examples

[Table 27](#) lists the applications of PrimeSim HSPICE exploration block demo files available in the `$installdir/demo/hspice/variability/` demo directory.

Table 27 PrimeSim HSPICE Exploration Block Demo Files

File name	Description	Reference to Detailed Description
EB_definition.txt	Testing matrix created from *.mex.csv file	See Exploration Block in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
EB_explore.sp	Case file to run design exploration	
EB_export.sp	Case file to generate design exploration variables	
readme.txt	Explains the design exploration phases	
snps_mac.lib	Model file	

Verilog-A Examples

[Table 28](#) lists the applications of PrimeSim HSPICE Verilog-A: netlist and Verilog-A demo files available in the `$installdir/demo/hspice/veriloga/` demo directory.

Table 28 PrimeSim HSPICE Verilog-A: Netlist and Verilog-A Files

File name	Description	Reference to Detailed Description
biterrrate.sp	Bit error rate counter	See Verilog-A (pVA) Messages in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> .
biterrrate.va		

Table 28 *PrimeSim HSPICE Verilog-A: Netlist and Verilog-A Files (Continued)*

File name	Description	Reference to Detailed Description
bjt.sp	BJT model	
bjt.va		
colpitts.sp	Colpitts BJT oscillator	
colpitts.va		
dac.sp	DAC and ADC	
dac.va		
deadband.sp	Deadband amplifier	
deadband.va		
ecl.sp	ECL inverter	
PrimeSim HSPICE_Veril og-A_Examples. txt	readme file for Verilog-A examples	
onehot.sp	Integer-to-one hot binary converter	
onehot.va		
opamp.sp	Opamp	
opamp.va		
pll.sp	Behavioral model of PLL	
pll.va		
resistor.sp	Very simple Verilog-A resistor model	
resistor.va		
sample_hold.sp	Sample and hold	
sample_hold.va		
sinev.sp	Simple voltage source	
sinev.va		

3

Demonstration Netlist Examples

Discusses in detail a few PrimeSim HSPICE demo files. Contains examples of basic netlist file construction techniques, demo objectives, display of results, and simulation hints.

This chapter discusses the following topics:

- [Two-Bit Adder Demo](#)
- [MOS I-V and C-V Plotting Demo](#)
- [CMOS Output Driver Demo](#)
- [Temperature Coefficients Demo](#)
- [Modeling Wide-Channel MOS Transistors Demo](#)

Two-Bit Adder Demo

Two-Bit Adder Demo file location: `$installdir/demo/hspice/apps/mos2bit.sp`

This two-bit adder shows how to improve efficiency, accuracy, and productivity in circuit simulation. It consists of two-input NAND gates, defined using the NAND subcircuit. CMOS devices include length, width, and output loading parameters. Descriptive names enhance the readability of this circuit.

One-Bit Subcircuit

The One-BIT subcircuit defines the two half adders, with carry in and carry out. To create the two-bit adder, PrimeSim HSPICE uses two calls to One-BIT. Independent piecewise linear voltage sources provide the input stimuli. The *R* repeat function creates complex waveforms.

Figure 2 Two-bit Adder Circuit

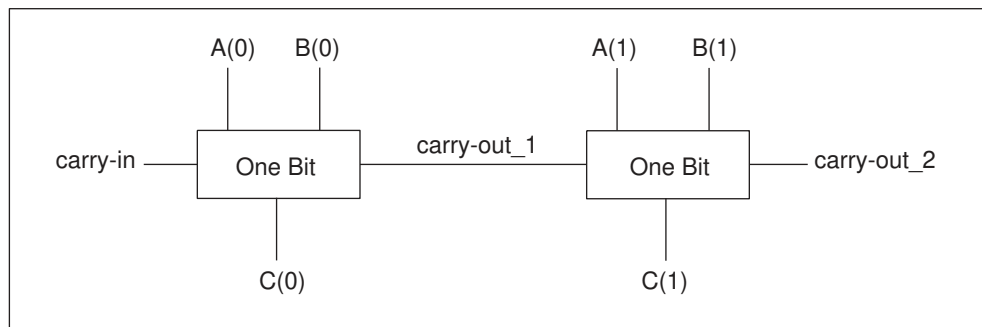
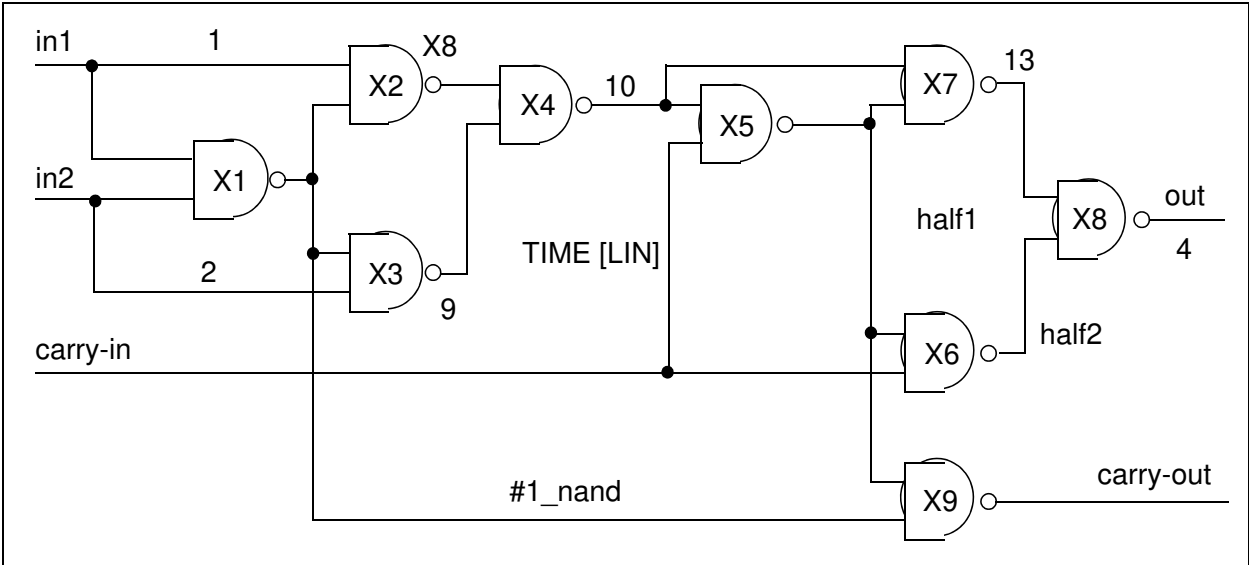


Figure 3 1-bit NAND Gate Binary Adder



MOS I-V and C-V Plotting Demo

MOS I-V and C-V plotting demo file location: `$installdir/demo/hspice/mos/mosivcv.sp`

To diagnose a simulation or modeling problem, you usually need to review the basic characteristics of the transistors. This example shows how to easily create input files, and how to display the complete graphical results. Table 27 lists the features that aid model evaluation.

You can use this demonstration template file with any MOS model.

Table 29 MOS I-V and C-V Plotting Demo

Value	Description
SCALE=1u	Sets the element units to microns (not meters). Most circuit designs use microns.
DCCAP	Forces PrimeSim HSPICE to evaluate the voltage variable capacitors, during a DC sweep.
node names	Eases circuit clarity. Symbolic name contains up to 16 characters.
.PRINT	.PRINT statements print internal variables.

Printing Variables

Table 30 lists the variables that the template provides for printing graphical results.

Table 30 Demo Printing Variables

Variable	Description
i(mn1)	i1, i2, i3, or i4 specifies true branch currents for each transistor node.
LV18(mn6)	Total gate capacitance (C-V plot).
LX7(mn1)	GM gate transconductance. (LX8 specifies GDS; LX9 specifies GMB).

Figure 4 MOS IDS Plot

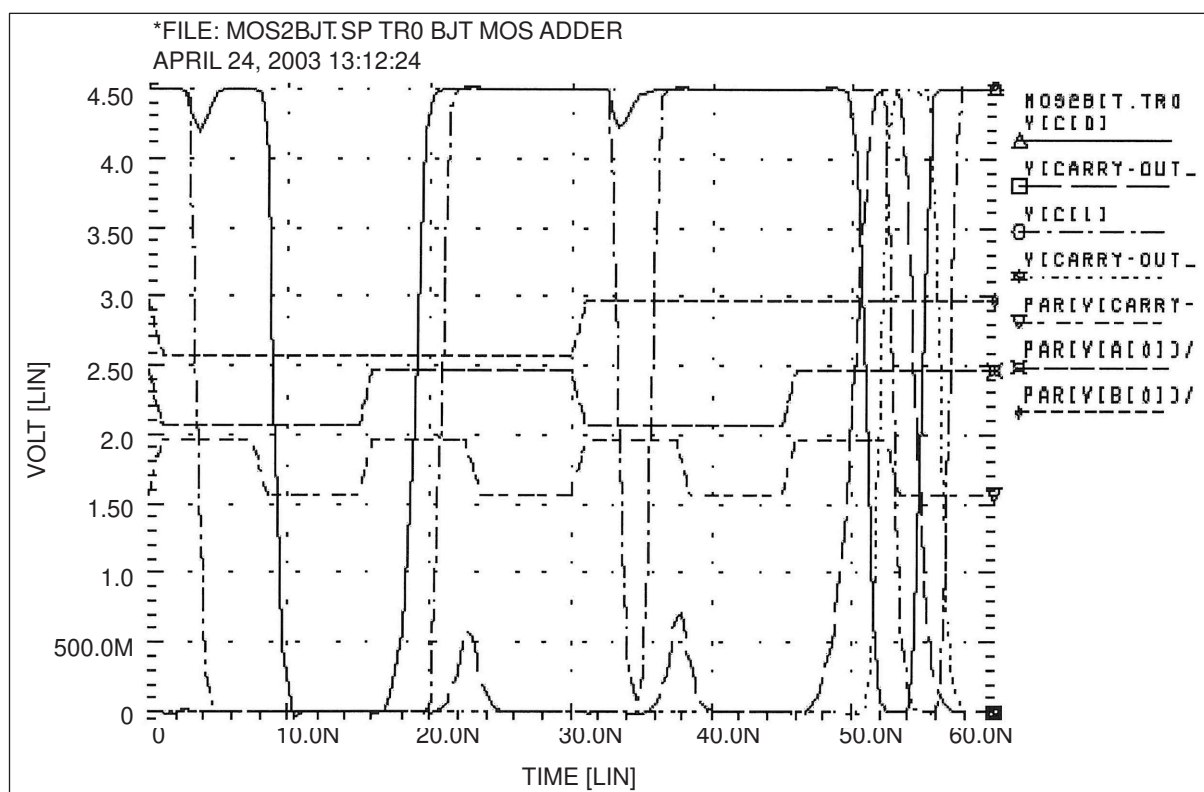


Figure 5 MOS VGS Plot

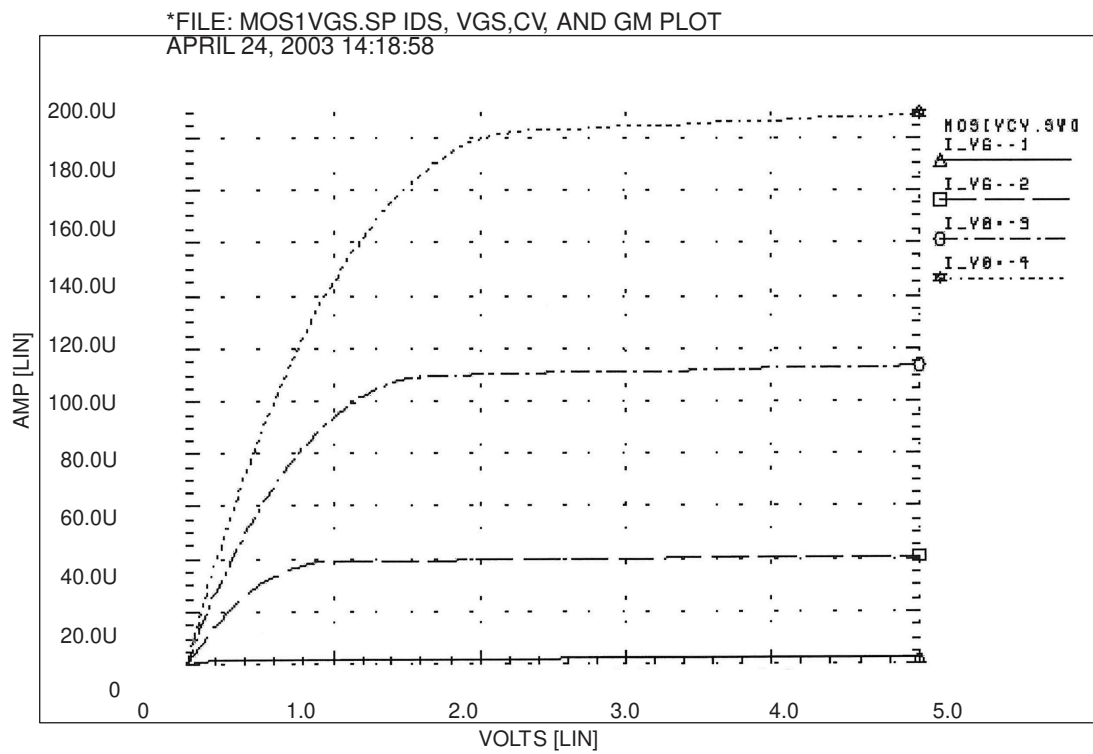


Figure 6 MOS GM Plot

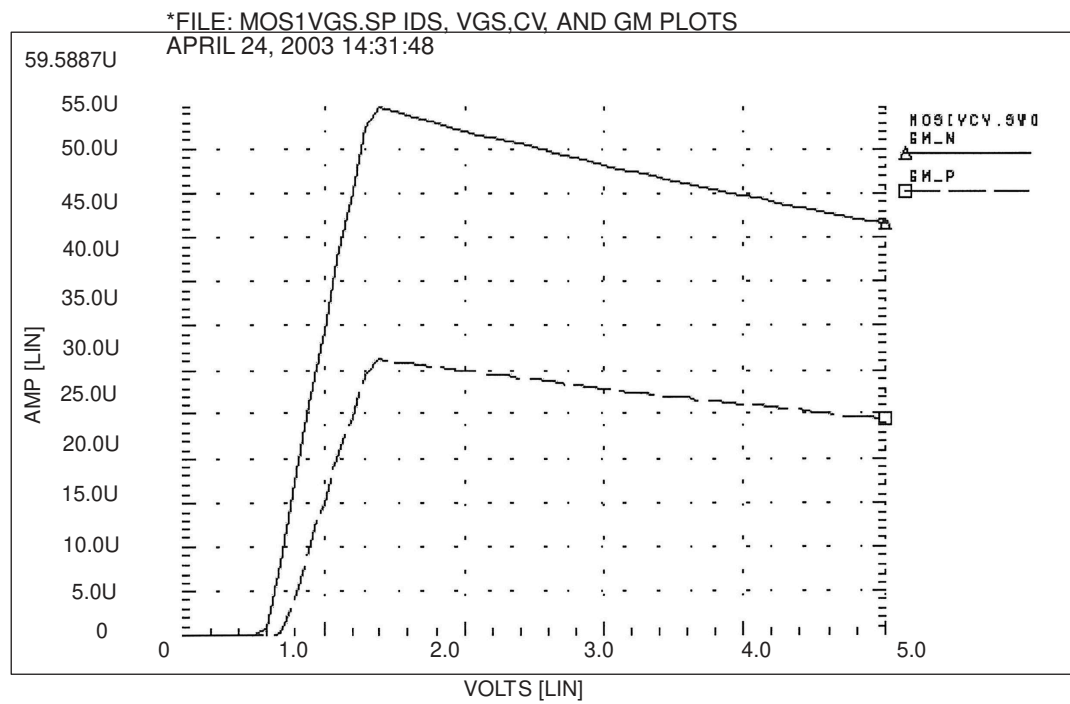
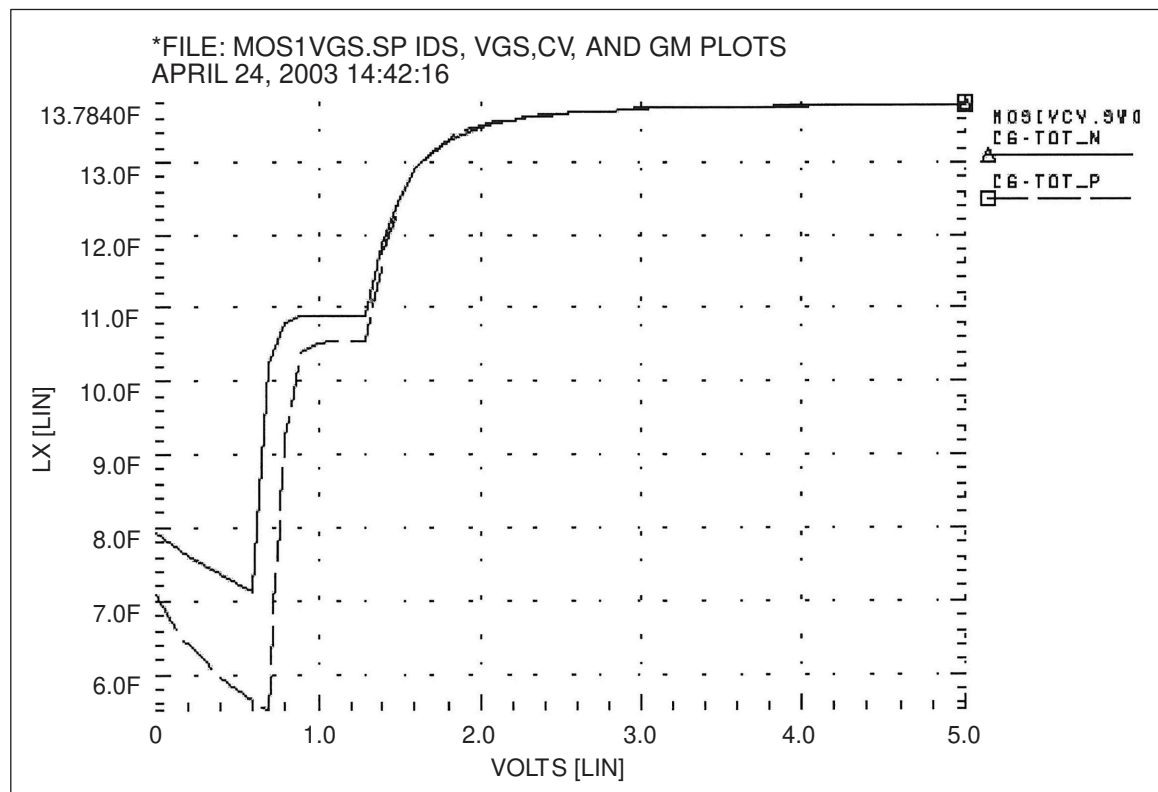


Figure 7 MOS C-V Plot



CMOS Output Driver Demo

CMOS output driver demo file location: `$installdir/demo/hspice/apps/asic1.sp`

ASIC designers need to integrate high-performance IC parts onto a printed circuit board (PCB). The output driver circuit is critical to system performance. The demonstration file shows models for an output driver, the bond wire and leadframe, and a six-inch length of copper transmission line.

This simulation demonstrates how to:

- Define parameters, and measure test outputs
- Use the LUMP5 macro to input geometric units, and convert them to electrical units
- Use `.MEASURE` statements to calculate the peak local supply current, voltage drop, and power
- Measure RMS power, delay, rise times, and fall times

- Simulate and measure an output driver under load. The load consists of
 - Bondwire and leadframe inductance
 - Bondwire and leadframe resistance
 - Leadframe capacitance
 - Six inches of 6-mil copper, on an FR-4 printed circuit board
 - Capacitive load, at the end of the copper wire

Strategy

The PrimeSim HSPICE strategy is to:

- Create a five-lump transmission line model for the copper wire
- Create single lumped models for leadframe loads

Figure 8 Noise Bounce

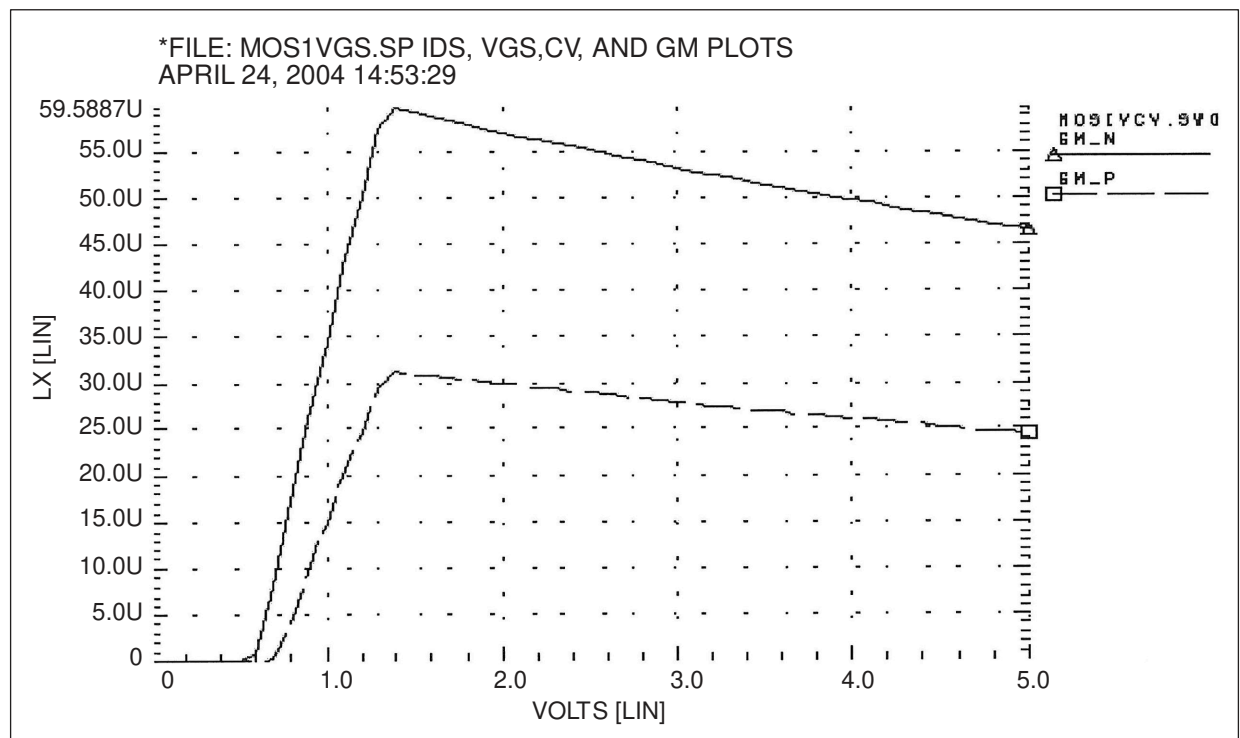


Figure 9 Asic1.sp Demo Local Supply Voltage

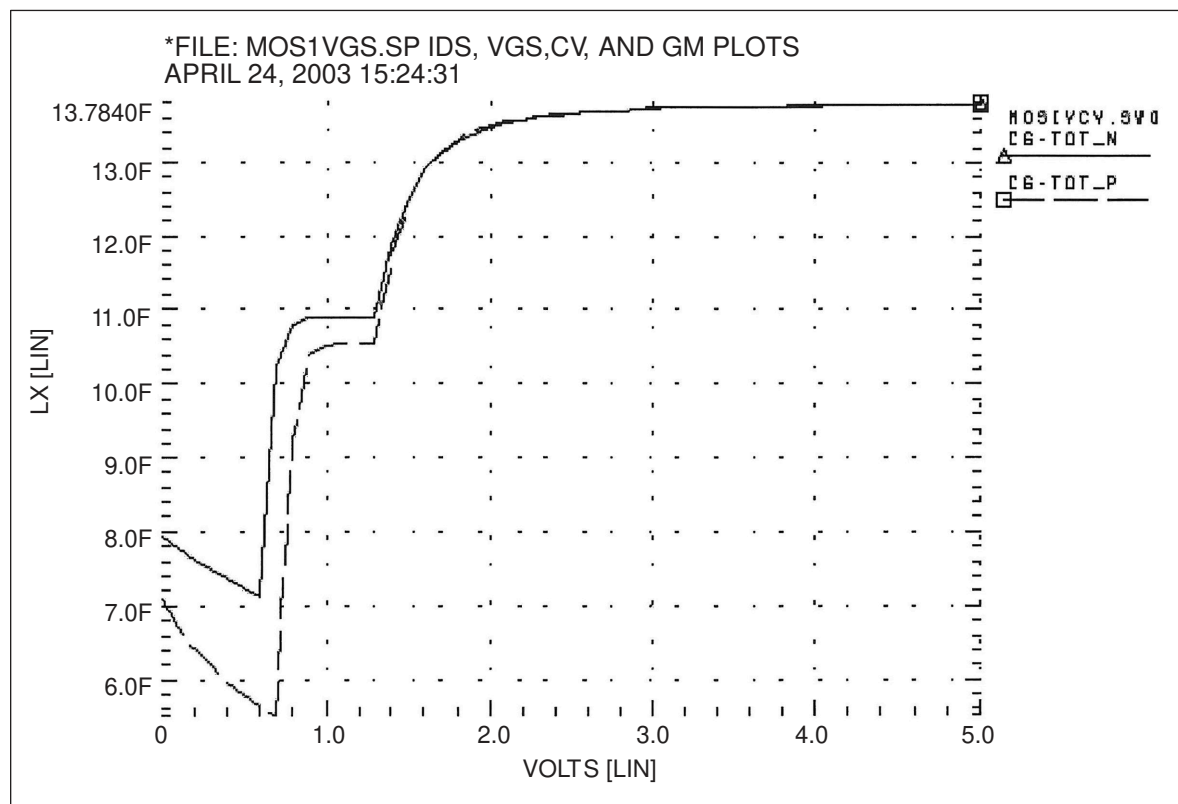


Figure 10 Asic1.sp Demo Local Supply Current

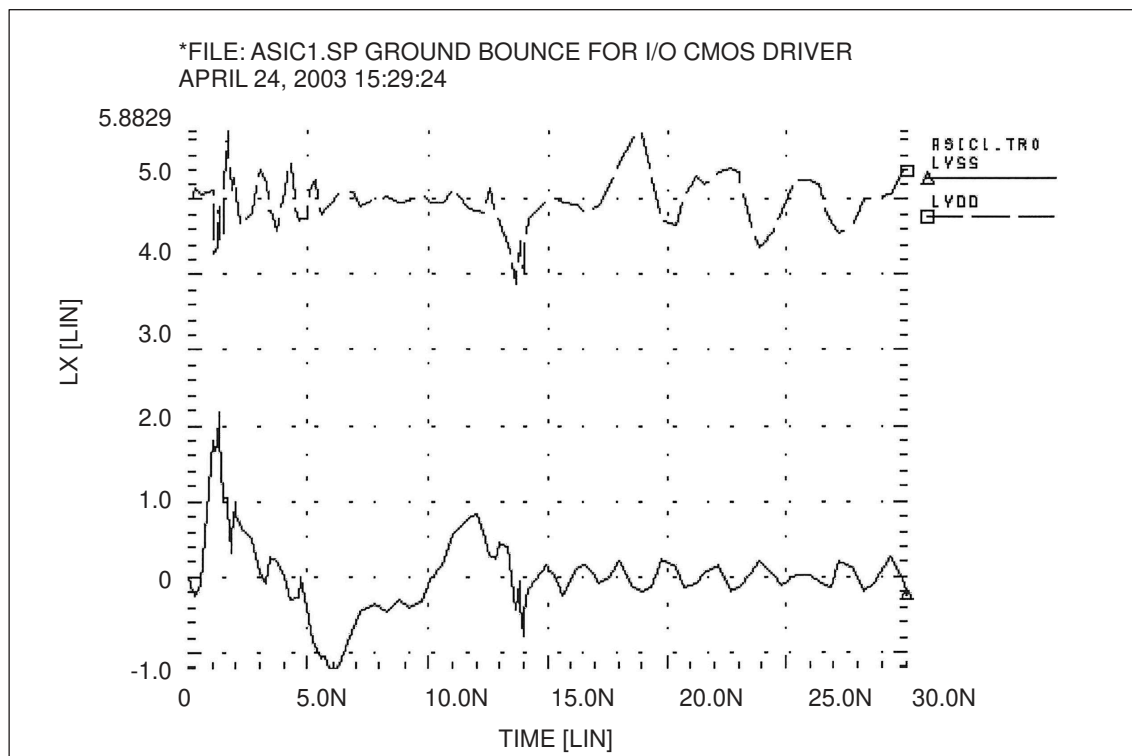
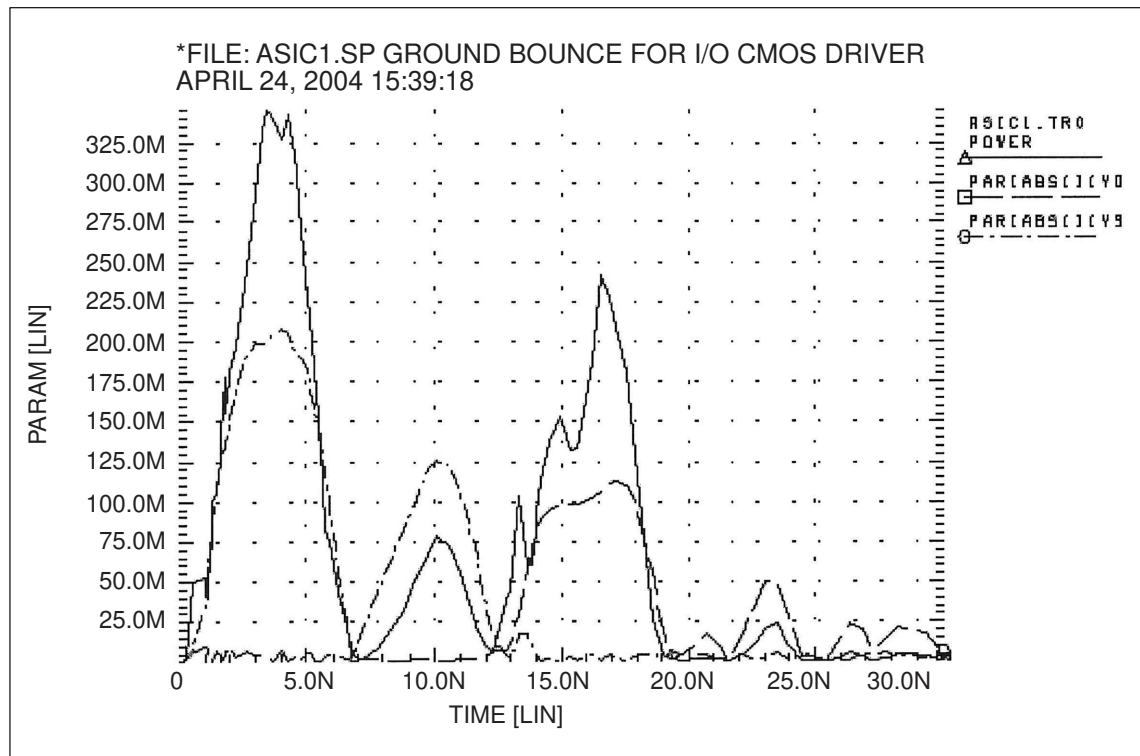


Figure 11 Asic1.sp Demo Input and Output Signals



Temperature Coefficients Demo

Input file for optimized temperature coefficients demo file location: `$installdir/demo/hspice/ciropt/opttemp.sp`

SPICE-type simulators do not always automatically compensate for variations in temperature. The simulators make many assumptions that are not valid for all technologies. Many of the critical model parameters in PrimeSim HSPICE provide first-order and second-order temperature coefficients, to ensure accurate simulations.

You can optimize these temperature coefficients in either of two ways.

- The first method uses the `TEMP` DC sweep variable.

All analysis sweeps allow two sweep variables. To optimize the temperature coefficients, one of these must be the optimize variable. Sweeping `TEMP` limits the component to a linear element, such as a resistor, inductor, or capacitor.

- The second method uses multiple components at different temperatures.

Example

The demo file, simulates three circuits of a voltage source. It also simulates a resistor at -25, 0, and +25°C from nominal, using the `DTEMP` parameter for element delta temperatures. The resistors share a common model.

You need three temperatures to solve a second-order equation. You can extend this simulation template to a transient simulation of non-linear components, such as bipolar transistors, diodes, and FETs.

This example uses some simulation shortcuts. In the internal output templates for resistors, LV1 (resistor) is the conductance (reciprocal resistance) at the desired temperature.

- You can run optimization in the resistance domain
- To optimize more complex elements, use the current or voltage domain, with measured sweep data

The error function expects a sweep on at least two points, so the data statement must include two duplicate points.

Optimization Section

```
.model optmod opt
.dc data=RES_TEMP optimize=opt1
+      results=r@temp1,r@temp2,r@temp3
+      model=optmod
.param tc1r_opt=opt1(.001,-.1,.1)
.param tc2r_opt=opt1(1u,-1m,1m)
.meas r@temp1 err2 par(R_meas_t1) par('1.0 / lv1(r-25)')
.meas r@temp2 err2 par(R_meas_t2) par('1.0 / lv1(r0) ')
.meas r@temp3 err2 par(R_meas_t3) par('1.0 / lv1(r+25) ')
* * Output section *
.dc data=RES_TEMP
.print 'r1_diff'=par('1.0/lv1(r-25)')
+      'r2_diff'=par('1.0/lv1(r0) ')
+      'r3_diff'=par('1.0/lv1(r+25) ')
.data RES_TEMP R_meas_t1 R_meas_t2 R_meas_t3
950 1000 1010
950 1000 1010
.enddata
.end
```

Modeling Wide-Channel MOS Transistors Demo

Nine-stage ladder model netlist file location: `$installdir/demo/hspice/apps/asic3.sp`.

If you select an appropriate model for I/O cell transistors, simulation accuracy improves. For wide-channel devices, model the transistor as a *group* of transistors, which are connected in parallel, with appropriate RC delay networks. If you model the device as only *one* transistor, which reduces the polysilicon gate introduces delay.

When you scale to higher-speed technologies, the area of the polysilicon gate decreases the gate capacitance. However, if you scale the gate oxide thickness, the capacitance per unit area increases, which also increases the RC product.

Example

The following example illustrates how scaling affects the delay. For example, for a device with:

- Channel width = 100 microns.
- Channel length = 5 microns.
- Gate oxide thickness = 800 Angstroms.

The resulting RC product for the polysilicon gate is:

$$R_{poly} = \frac{W}{L} \cdot 40 \quad \text{poly} = \frac{E_{si} \cdot n_{si}}{t_{ox}} \cdot L \cdot W$$

$$R_{poly} = \frac{100}{5} \cdot 40 = 800, \quad C_o = \frac{3.9 \cdot 8.86}{800} \cdot 100 \cdot 5 = 215 fF \quad RC = 138 \text{ ps}$$

For a transistor with:

- Channel width = 100 microns.
- Channel length = 1.2 microns.
- Gate oxide thickness = 250 Angstroms.

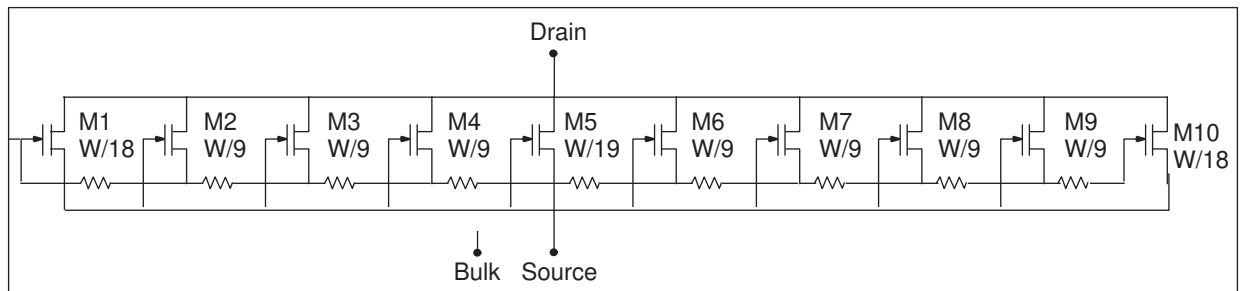
The resulting RC product for the polysilicon gate is:

$$R_{poly} = \frac{\text{channelwidth}}{\text{channellength}} \cdot 40$$

$$C_o = \frac{3.9 \cdot 8.86}{T_{ox}} \cdot \text{channelwidth} \cdot \text{channellength} \quad RC = 546 \text{ ps}$$

Figure 12 shows how to use a nine-stage ladder model to model the RC delay in CMOS devices.

Figure 12 Nine-stage Ladder Model



In this example, the nine-stage ladder model is in data file `$installdir/demo/hspice/apps/asic3.sp`. To optimize this model, PrimeSim HSPICE uses measured data from a wide channel transistor as the target data. Optimization produces a nine-stage ladder model, which matches the timing characteristics of the physical data. The PrimeSim HSPICE advanced analog analyses does not support optimization. PrimeSim HSPICE compares the simulation results for the nine-stage ladder model with the one-stage model by using the nine-stage ladder model as the reference. The one-stage model results are about 10% faster than actual physical data indicates.

Input File

You can find the sample Nine-Stage Ladder model netlist for this example in the following directory:

```
$installdir/demo/hspice/apps/asic3.sp
```

Figure 13 Asic3 Single vs. Lumped Model

